

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (Mechanical) (Credit System) Degree Examinations

Supplementary - April 2023

20ME305 / 17ME305 – MECHANICS OF MATERIALS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Define the following. (i) Shear stress (ii) Lateral Strain (iii) Poisson's ratio (iv) stiffness (v) Plasticity | 10 | L*2 | 1 | 1 |
| b) The following data refers to the tensile test conducted on a mild steel bar.
i. Diameter of the steel bar = 25 mm
ii. Gauge length = 300 mm
iii. Extension at a load of 15kN = 0.045 mm
iv. Load at yield = 127.65kN
v. Maximum load = 208.6kN
vi. Length of specimen at failure = 375mm
vii. Neck diameter = 17.75mm
Calculate I) Young's modulus II) stress at yield point III) The percentage of elongation IV) ultimate stress V) percentage reduction in area. | 10 | L3 | 1 | 2 |
| 2. a) Derive an expression for the extension of a tapering rectangular bar of cross section varies from b_1 and b_2 over a length l when it is subjected to an axial pull P and Young's modulus E . | 6 | L2 | 1 | 2 |
| b) Explain the plane stress and plane strain conditions with suitable example. | 4 | L2 | 1 | 3 |
| c) A member ABCD is subjected to point loads P_1 , P_2 and P_3 as shown in Figure Q2(b). Determine the total elongation of the member assuming the modulus of elasticity to be 200GPa and cross section area $A = 225 \text{ mm}^2$ | | | | |



Figure Q2(b)

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 3. a) Derive an expression for relationship between Young's modulus, modulus of rigidity and poisson's ratio. | 10 | L2 | 2 | 1 |
| b) A steel rod of 4m long 20 mm diameter is subjected to an axial tensile load of 40kN. Determine the change in length, diameter and volume of rod. $E = 200\text{GPa}$ and poisons ratio $= 0.25$. | 10 | L3 | 2 | 2 |

4. a) State the assumptions made in Euler's column theory.
 b) Derive an expression for Hoop stress and longitudinal stress for thin cylinders.
 c) A thick cylinder pipe of outside diameter 300 mm and internal diameter 200 mm is subjected to an internal pressure of 20 N/mm² when the external pressure of 5 N/mm². Determine the maximum hoop stress developed. Draw the variation of hoop stress and radial stress across the thickness indicating the values at every 25 mm interval.

4	L2	2	1
6	L2	2	1
10	L3	2	2

Unit – III

5. Find the reaction forces at A and B and draw the shear force and bending moment diagram for the beam shown in Figure Q5(a). Also find maximum bending moment.

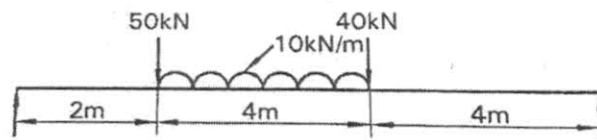


Figure Q5(a).

20	L3	3	2
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6. For the beam shown in Figure Q6(a) determine the magnitude of the load P acting at C, such that the reaction at supports A and B are equal. Draw the shear force and bending moment diagrams. Locate the point of contra-flexure.

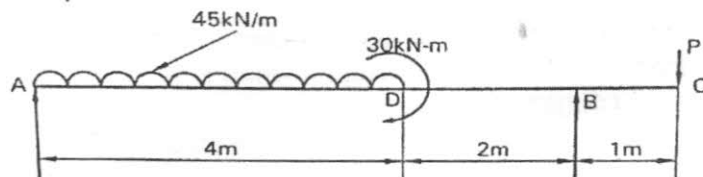


Figure Q6(a)

20	L3	3	2
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Unit – IV

7. a) Derive an expression for the bending stress and radius of curvature for a straight beam subjected to pure bending. Also state the assumptions made in the theory of simple bending.
 b) A beam of T section has a length of 2.5 m and is subjected to a point load as shown in Figure Q7(b). Calculate the compressive bending stress and plot the stress distribution across the cross-section of the beam. The maximum tensile stress is limited to 300MPa. Calculate value of W

10	L3	4	2
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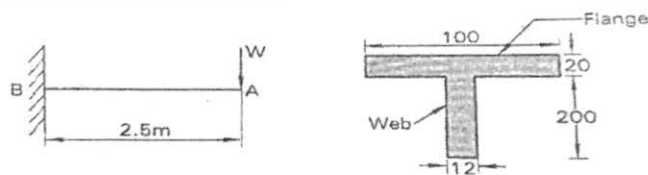


Figure Q7(b)

10	L3	4	2
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8. a) An I section has flange of width b and over all depth 2b. The flanges and web are of uniform thickness t. Find the ratio of maximum shear stress intensity to the mean shear intensity.
 b) A beam of T-section has flanges 100 mm x 20 mm and web 200 mm x 12 mm is subjected to vertical shear force of 200kN. Find the shear stress at the flange, junction and neutral axis. Also sketch the stress distribution.

10	L3	4	2
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10	L3	4	2
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Unit – V

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|--------|--|----|----|---|---|
| 9. a) | Derive an expression with usual notations for the maximum deflection in a cantilever beam subjected to point load at the free end. | 10 | L3 | 5 | 2 |
| b) | A simply supported beam having uniform cross-section is 14 m span and is simply supported at its end. It carries a concentrated load of 120kN and 80kN at two points 3 m and 4.5 m from the left and right end respectively. If the moment of inertia of the section is $160 \times 10^7 \text{ mm}^4$ and $E = 210\text{GPa}$, calculate the deflection of the beam at load points and mid span. | 10 | L3 | 5 | 2 |
| 10. a) | Compare the weight of solid shaft with that of hollow one having the same length to transmit a given power at a given speed if the material used for both shafts is the same. The inside diameter of hollow shaft is 0.6 times outer diameter. | 10 | L3 | 5 | 2 |
| b) | A solid shaft rotating at 1000 rpm transmits 50kW. Maximum torque is 20% more than mean torque. Material of the shaft has the allowable shear stress of 50MPa and modulus of rigidity 80GPa. Angle of twist in the shaft should not exceed 1° in one meter length. Determine the diameter of the shaft. | 10 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
